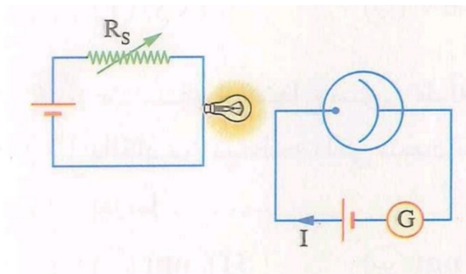
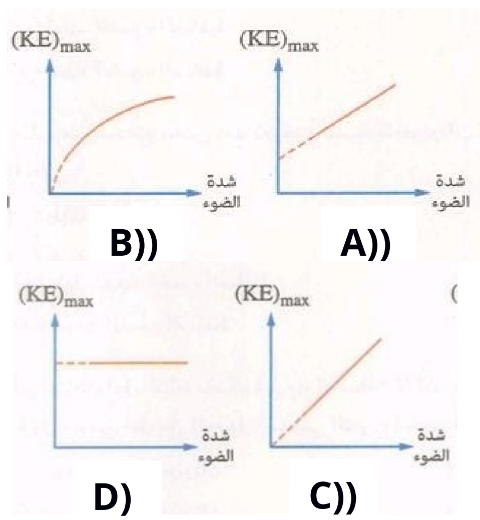
الكود	المعرفة
TPK01	Photoelectric effect

1-Choose the correct answer:

#	Question	Choices	Answer
1	The process in which electrons are emitted from a metallic surface when a light given to it is called	a. Photoelectric effect b. Thermionic effect c. Compton effect d. Black body radiation	a
2	the emission of electrons from metallic surface (photoelectric effect) depends on	a. The intensity of the incident light b. The frequency of incident light c. The exposure time	b
3	The process in which electrons are emitted from a metallic surface when a light given to it is called	a. Photoelectric effect b. Thermionic effect c. Compton effect d. Black body radiation	a
4	The emission of electrons from cathode in photoelectric cell is depend on ……	a. The type of material of Anode and the light intensity. b. The type of material of Anode and the light frequency. c. The type of material of Cathode and the light intensity. d. The type of material of Cathode and the light frequency.	d

5	At the opposite figure shows light produced from an electric lamp falls on a photoelectric cell cases flow of photocurrent, if the light intensity increases, the photocurrent .... 	a. Increases. b. Decreases. c. Vanishes. d. Doesn't change.	d
6	Which of the following graphs represents the relationship between $K.E_{Max}$ for emitted photoelectrons and light intensity		d
7	An electromagnetic radiation is incident on a photoelectric cell with υ , photoelectrons emitted with $K.E_{Max}$, if the frequency becomes 2 υ, which of the following doesn't change	a. The photon energy. b. Speed of photon c. $K.E_{Max}$ for emitted photoelectron. d. Maximum velocity of emitted photoelectron.	b

8	A monochromic light incident on photoelectric cell with frequency greater than the critical frequency for cathode, then another light is used with the same wavelength of the first one but its intensity is double of the first, which one is doubled …..	a. The energy of each photon e. $K.E_{Max}$ for emitted photoelectron. b. Work function for the metal. c. The intensity of photocurrent.	C Because the frequency of incident light greater than E_w , so the intensity of photoelectrons is directly with the light intensity.
9	If the frequency of incident light increase to double , so the work function …..	a. Increase to double b. Decrease to half c. Remains constant d. Increase for four times.	c
10	When the incident light has wavelength 623 nm on a metal , photoelectrons are emitted with 4.6 x 10 ^{5} m/s, so : A- The work function of the metal ….. B- The critical frequency of the metal …….	(A) a. $2.23 \times 10^{-19} J$. b. $3.19 \times 10^{-19} J$. c. $4.15 \times 10^{-19} J$. d. $4.6 \times 10^{-19} J$. (B) a. 6.94×10^{14} b. $6.26 \times 10^{14} HZ$. c. $4.81 \times 10^{14} HZ$. d. $3.34 \times 10^{14} HZ$.	(A) b (B) c
11	If the critical frequency of a metal equals the frequency of the blue light , which of	a. Infrared rays. b. Radio waves. c. Red light.	d

	the following rays will be able to emit the electrons from the metal surface…..	d. Ultraviolet rays.	
12	If a light with 3100 Å incident on the surface of cathode in photoelectric cell, so the electrons are emitted with kinetic energy equals 2.5 eV , so the work function for the cathode is …….	a. 3.1 eV. b. 2.4 eV. c. 1.5 eV. d. 0.9 eV.	b
13	If the work function for a metal equals 3.3125×10^{-19} J , the critical frequency is ……	a. $4.5\times10^{14}\,HZ$. b. $4.8\times10^{14}\,HZ$. c. $5\times10^{14}\,HZ$. d. $5.5\times10^{14}\,HZ$.	c

2- Mention the factor affecting on

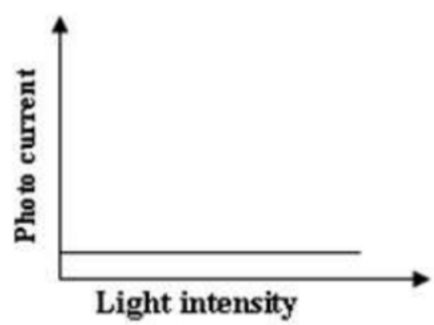
No.	Question	Answer
1	KE of the freed electron by photoelectric effect	Incident photon energy or Work function or frequency of light used
2	the velocity of the freed electron by photoelectric effect	Kinetic energy

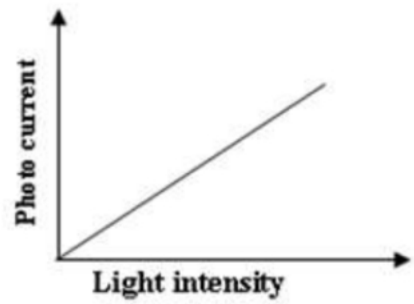
3- True or False

No.	Question	Answer	Correction if F
1	Changing the color of the light means we are changing the frequency.	T	
2	In Photoelectric effect, electrical energy converted to light energy	F	Light to electrical
3	Photoelectrons are electrons that are released or ejected from a substance by photoelectric effect	T	
4	Kinetic energy is independent of the intensity of light.	T	
5	Increasing the brightness of the light, increases the speed of the electrons.	F	independent
6	Increasing the intensity of the light, increases the number of electrons.	T	
7	increasing the frequency, increases the kinetic energy.	T	
8	Photoelectric effect is always instantaneous	T	
9	The photon energy doesn't have to be more than the work function for electrons to be ejected.	F	It has to be greater

4-Solve the following:

No.	Question	Answer
1	a monochromatic radiation of wavelength $3*10^{-7}$ m falls on a metal surface, so electrons liberated with maximum kinetic energy $3.26*10^{-19}$ J	$K.E_{max} = h\nu - E_w$ a) Photon energy = hv = h*$\frac{c}{\lambda}$ $6.63*10^{-19}$ J b) E_w = hv – KE

	a) the energy of the photon that falls on the metal surface b) the work function of the metal	$3.37 10^{-19}$ J
2	photon with wavelength $4*10^{-7}$ m falls on a surface of metal with work function $2.3 *10^{-19}$ J, find the kinetic energy of emitted electron from a surface of the metal	$K.E_{max} = h\nu - E_w$ $KE = \frac{h * 3 * 10^{8}}{4 * 10^{-7}} - 2.3 * 10^{-19}$ $2.67 10^{-19} J$
3	A metal has a work function 8 x 10^{-19} J. Calculate the maximum kinetic energy of electrons emitted from the metal when light of frequency 2 ×10^{15} Hz is shone on the surface.	① $E_w = 8*10^{-19}$ J $f = 2*10^{15}$ Hz $KE = hf - E_w = 6.625*10^{-34}*2*10^{15} - 8*10^{-19}$ $= 5.25*10^{-19}$ J
4	When monochlight of wave length 663 nm is incident on a metal plate whose work function 3.52 ×10^{-19} eV will electrons be emitted or not?	② $\lambda = 663 nm \longrightarrow f = \frac{c}{\lambda} = \frac{3*10^{8}}{663*10^{-9}} = 4.525*10^{14}$ Hz $E_w = 3.52*10^{-19} eV \longrightarrow 3.52*10^{-19} * e = 5.64*10^{-38}$ J $E = hf = 2.998*10^{-19}$ J $E > E_w$ ∴ emitted
5	Draw the relationship between light intensity and photocurrent at the case of $E < E_W$ and $E \geq E_W$.	$E < E_W$  (a) (b) $E \geq E_W$

5- complete the following sentences:

No.	Question	Answer
1	There are many ways to eject (escape) electrons out of a material are ……………………… and …………………..	Thermionic emission – photoelectric effect
2	The emission of electrons from a metallic surface when the light energy falling on is ……………………………………	Photoelectric effect
3	The electrons that released or ejected from a substance by photoelectric effect is called ……………………………… .	Photoelectrons
4	An application on photoelectric effect is …………….. .	Photoelectric cell
5	photoelectric cell aims to convert ………………………………… To ………………………………… .	Light energy-electric energy
6	the source of free electrons in photoelectric cell is ……………………….. .	Cathode
7	the collector in photoelectric cell is called ………………………………… .	Anode
8	the minimum energy needed to free (release) the electrons from the surface of	Work function

	the metal is	
z	To free (emit) an electron with kinetic energy from the metal, the energy of the incident photon must be.............................. .	Electron gun
10	The kinetic energy of the emitted electrons is independent of	Fluorescent screen
12	At the case of .... the relationship between light intensity and photocurrent is independent of each other.	EE_w
13	At the case of .... the relationship between light intensity and photocurrent is directly proportional to each other.	EE_w

5- Mention the scientific term

No.	Question	Answer
1	An electron released or ejected from a substance by photoelectric effect	Photoelectrons
2	It is the minimum energy needed to free (release) the electrons from the surface of the metal	work function of the metal